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$$v'(x) = \begin{cases} \alpha - \gamma x & x \geq 0 \\ \lambda(\alpha + \gamma x) & x < 0 \end{cases}$$

$$v'(y-r) \text{ vs } p_1 v'(y+\delta-r) + p_2 \lambda v'(-(y-\delta-r))$$

$$\hookrightarrow y-r = x$$

$$v'(x) \text{ vs } p_1 v'(x+\delta) + p_2 \lambda v'(\delta-x)$$

$$V_t(a_t, h_t, z_t, x_t, i_t) = \max_{c_t, l_t, i_{t+1}} \{v(c_t, l_t) + \dots$$

$$\beta \sum_{x_{t+1}} \sum_{h_{t+1}} \sum_{z_{t+1}} T(z_{t+1} | z_t) \psi^h(x_{t+1} | x_t) G(h_{t+1} | h_t)$$

$$\cdot V_{t+1}(a_{t+1}, h_{t+1}, x_{t+1}, z_{t+1}, i_{t+1})$$

Indicator function:

$$1_{\{s.t.\}} = \begin{cases} 1, & \text{if statement is true} \\ 0, & \text{else} \end{cases}$$

Budget Constraint : young

$$c_t + a_{t+1} + x_t (1 - q \cdot \mathbb{1}_{\{i_t=1\}}) + p(x_t, h_t) \mathbb{1}_{\{i_{t+1}=1\}} + \tau x_t = w z_t^h l_t + a_t (1+r)$$

Budget Constraint : retired

↳ no earning, receive pensions pens

↳ all covered medicare, covering $(1-q)$

$$c_t + a_{t+1} + x_t (1-q) + \tau x_t = \text{pens} + a_t (1+r)$$

$$V_t^R(h_t, c_t, a_t, x_t) = \max_{\{c_{t+1}, a_{t+1}\}} \{ U(c_t, 0) + \beta S^t \sum_{x_{t+1}} \sum_{h_{t+1}} \psi^n(x_{t+1} | x_t) \}$$

$$G(h_{t+1} | h_t) \cdot V_{t+1}^R(h_{t+1}, a_{t+1}, c_{t+1}, x_{t+1})$$

Value Function Iteration:

$$v(k) = \max_{c, k'} \{u(c) + \beta v(k')\}$$

s.t.

$$c + k' = z f(k) + (1 - \delta)k$$

↳ plug in for k' s.t. $v(0) \xrightarrow{v(k')} v(1) \xrightarrow{v(k)} v(n)$

$$\text{s.t. } \|v^{(n+1)} - v^{(n)}\| < \text{tol.}$$

↳ discretize state space

$$\{\underbrace{k}_k, \dots, \underbrace{k_n}_k\}$$

→ at k close to $\underline{k}$, many points $\Rightarrow$ space as you go towards $\bar{k}$

↳ Algorithm:

- ① choose err. tolerance level ε ($\sim 10^{-6}$)
- ② discretize state space $\rightarrow \{k, \dots, k_n\}$
↳ O w/ k for k
- ③ guess $v^0(k)$ ($1 \times n$ vector)
- ④ update guess

VFI: Step 4:

$$v^{(1)}(k_i) = \max_{k'} \{ u(zf(k_i) + k_i(1-\delta) - k') + \beta v^{(0)}(k') \}$$

$$\hookrightarrow T_{ij} = u(zf(k_i) + k_i(1-\delta) - k_j) + \beta v^0(k_j)$$

$$T = \begin{matrix} & \begin{matrix} k_1 & k_2 & \text{Tomorrow} & & k_j \end{matrix} \\ \begin{matrix} \text{Today} \\ k_1 \\ \vdots \\ k_i \\ \vdots \\ k_c \end{matrix} & \begin{bmatrix} T_{11} & T_{12} & \cdot & \cdot & \cdot & \cdot & k_{1j} \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ T_{i2} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \vdots & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \vdots & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \vdots & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ k_{i1} & \cdot & \cdot & \cdot & \cdot & \cdot & k_{ij} \end{bmatrix} \end{matrix}$$

Pick max T for each row

$\max_j T_{ij}$ for each i

$$\operatorname{argmax}_j T_{ij} = j^* \Rightarrow v_i^1 = \max_j T_{ij}^*$$

$$p(k_i) = k_{j^*}$$

Assume $v_1^1 = T_{12}$, $j^* = 2$

$$\Rightarrow v_2^1 = T_{13}, j^* = 3$$

updated vector

$$v^{(1)} = \begin{bmatrix} v_1^{(1)} & v_2^{(1)} & \dots & v_n^{(1)} \end{bmatrix}$$

$\parallel$ $\parallel$ $\parallel$
 $\max_j T_{1j}$ $\max_j T_{2j}$ $\max_j T_{nj}$

VFI Step 5:

Compare $v^{(0)}$ and $v^{(1)}$

$$\text{dist} = \max |v_i^{(1)} - v_i^{(0)}|$$

if $\text{dist} > \text{tol} \Rightarrow \text{repeat 4}$

$$4: T_{ij} = U(\alpha f(k_i) + (1-\delta)k_i - k_j) + \beta v^{(1)}(k_j)$$

$$\text{dist} = \max |v_i^{(2)} - v_i^{(1)}|$$

if $\text{dist} > \text{tol} \Rightarrow \text{repeat 4}$

else $\Rightarrow$ converged

$$k' = p^{(n+1)}(k)$$